

Class Imbalance in Histopathology: Replacing Virchow2 with UNI2-h Lowers TCGA-UT’s Damage and Leaves BRACS Unresolved

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Abstract

Earlier studies measured how much balanced accuracy a linear probe on frozen Virchow2 features loses when its training allocation is long-tailed, and split that damage into a prior channel, a support channel, and their interaction. This report asks whether those results transfer to a second encoder. On identical patients and image patches of TCGA-UT (30 classes) and BRACS (7 classes), frozen Virchow2 and frozen UNI2-h each train seven paired probes per split and draw, balanced (B), prior-only, support-only, and combined (R) at imbalance ratios 10 and 100, 840 probes in total, with validation-selected regularization per arm. The primary contrast is the encoder difference in the combined damage at ratio 100, judged with 97.5% paired-bootstrap intervals against a one-point threshold. On TCGA-UT, UNI2-h is 1.78 points more accurate under balanced training and loses 0.86 points less at ratio 100 (interval -1.33 to -0.40): the direction is resolved, the practical size is not. On BRACS the contrast is -0.67 points (-2.75 to 1.85), unresolved, and both encoders collapse the tail. At ratio 10 the damage is nearly equal on both datasets. UNI2-h carries less support-only damage on both datasets, the only contrast that keeps its sign when each encoder’s regularization is held at its balanced value; under that fixed rule the total damage reverses and becomes larger for UNI2-h, so the tuned reduction depends on the probe re-selecting its penalty. The results describe two complete encoder-plus-probe pipelines, not foundation models in general, and identify no representational mechanism.

1 Motivation

Every class-imbalance study in this series so far used one frozen encoder, Virchow2 (Zimmermann et al., 2024). On that encoder, training a linear probe on an exponential long-tailed allocation with imbalance ratio $\rho = 100$ costs 2.91 points of test patient-macro balanced accuracy on TCGA-UT and 7.65 points on BRACS, with every patient identity and the total patch budget held fixed (Queisler, 2026g,f). Two follow-up studies split that damage into a prior-only part, which reweights the balanced rows toward the long-tailed class shares, and a support-only part, which trains on the long-tailed rows under a uniform class prior. On TCGA-UT the 2.91 points divided into 1.20 (prior), 0.53 (support), and a positive interaction of 1.18 points; on BRACS the 7.65 points divided into 4.47 and 3.29 points with an interaction of -0.11 points whose interval did not resolve it (Queisler, 2026b,a).

These numbers describe one pipeline: one encoder, its image transform and pooling, and one tuned linear readout. Whether they describe class imbalance in general or a property of Virchow2’s feature geometry is untested. The attempts to explain them from inside that geometry have not settled the question either. A separation-and-centre-correction rescue of BRACS depended on the regularization strength and failed its integrity and robustness gates (Queisler, 2026c). A patch-coverage intervention failed its precheck (Queisler, 2026h), and a test-set oracle for the regularization strength did not close the BRACS–TCGA-UT prior gap

(Queisler, 2026d). Earlier patient-shortage work showed that broad, independent patient support matters under frozen Virchow2 (Queisler, 2026e); this study holds the patient identities fixed within each dataset, so patient breadth cannot explain an encoder contrast.

This report therefore replaces the encoder and keeps everything else. On the same patients and the same image patches, it asks whether replacing frozen Virchow2 with frozen UNI2-h (Mahmood Lab, 2025) changes the damage that class imbalance does to balanced accuracy, and if so, whether the change arises in the prior channel, the support channel, or their interaction. A smaller damage estimate alone would be ambiguous, because a worse balanced baseline leaves less accuracy to lose. The report therefore states absolute balanced and imbalanced accuracy next to every damage contrast.

2 Design

The design reuses the three patient-disjoint splits and the frozen allocator of the earlier prevalence studies (Queisler, 2026g,f). TCGA-UT keeps its 30 cancer types, $G = 20$ training patients per class, and a total budget of $T = 19,200$ patches (Komura et al., 2022); BRACS keeps its seven lesion subtypes, $G = 10$ patients per class, and $T = 2,240$ patches (Brancati et al., 2022). The two budgets differ because only ten to thirteen BRACS training patients per class reach the eligibility depth of 160 patches. Primary inference is therefore within each dataset; a comparison of the two datasets is descriptive and cannot isolate tissue type or class count. Ten fresh training draws per split (draw identifiers 10 to 19, unused by any earlier study) give 30 fits per encoder and arm. The evaluation patients are the same as in the earlier studies, and those studies informed this follow-up, so the fresh draws renew the training samples but not the test cohort.

2.1 Encoders and paired inputs

The patient cohort, the class-rank permutation, and the exact training patch identities of every draw were generated once, without reference to either encoder, and each encoder consumed the same schedule. Virchow2 features were reused from the audited existing caches: a 2,560-dimensional vector that concatenates the class token with the mean patch token. UNI2-h features were extracted from the pinned checkpoint of its model card, with the card’s timm architecture, the recommended 224-pixel resize-and-centre-crop transform with ImageNet normalization, and its single pooled 1,536-dimensional token, in float32 (Mahmood Lab, 2025). UNI2-h extends the UNI model family (Chen et al., 2024); its checkpoint is documented by the model card, not by the UNI paper. The feature audit resolved all 1,201,809 requested TCGA-UT patches and all 168,585 BRACS patches, row for row, against the shared manifest. The comparison concerns each complete pipeline: frozen encoder, its recommended transform and pooling, and the tuned probe on its native features, without PCA, whitening, or normalization.

2.2 Four-arm decomposition

For each encoder, split, and draw, seven multinomial logistic probes were fitted. The balanced arm B trains on the balanced allocation. The ratio arm $R\rho$, $\rho \in \{10, 100\}$, trains on the long-tailed allocation with its own realized class prior. The prior arm $P\rho$ keeps B’s rows and reweights the loss toward $R\rho$ ’s realized class shares, and the support arm $S\rho$ keeps $R\rho$ ’s rows and reweights the loss back to uniform class shares. Row weights are the ratio of target to observed class share and sum to the number of rows. $S\rho$ therefore measures a redistribution of patch support at a fixed total budget, not the removal of tail rows alone. With $A(m, a)$ the test patient-macro balanced accuracy of encoder m and arm a , averaged over draws within a split and then equally

over the three splits,

$$D_X(m, \rho) = A(m, B) - A(m, X\rho), \quad X \in \{P, S, R\}, \quad I(m, \rho) = D_R - D_P - D_S, \quad (1)$$

so positive damage means worse than balanced training. The primary contrast, one per dataset, is

$$\delta_R = D_R(\text{UNI2-h}, 100) - D_R(\text{Virchow2}, 100), \quad (2)$$

where a negative value means that UNI2-h loses less accuracy under imbalance. The component contrasts δ_P , δ_S , and δ_I are defined the same way and sum exactly to δ_R . Ratio 10 is a secondary severity check; two ratios cannot identify the shape of a damage curve.

2.3 Readout, selection, and calibration

Every encoder and arm received the same selection procedure, not the same selected value: an L2 penalty λ chosen from eleven decades $10^{-8}, \dots, 10^2$ by validation patient-macro balanced accuracy, with ties broken toward stronger regularization. The grid was frozen before the main run; Section 4.2 records how its boundary rule was applied. As a secondary sensitivity, each encoder’s P100, S100, and R100 arms were re-evaluated at that encoder’s own B-selected λ , using stored candidate coefficients and no refit. The two encoders’ numerical λ values are never exchanged, because their feature scales differ. One positive temperature per selected arm was fitted on validation patients; it leaves the argmax, and therefore balanced accuracy, unchanged (Guo et al., 2017). Probability quality is reported as the macro negative log-likelihood (NLL), which weights every test patient and class equally and depends on discrimination as well as calibration, and as the expected calibration error (ECE) in ten fixed top-label confidence bins with patient weights. ECE depends on the evaluation prevalence, which is the same for all arms and encoders, and it is not class-balanced.

2.4 Inference and decision rule

Intervals come from 10,000 paired bootstrap replicates with seed 39000. Each replicate resamples training draws within split with identical weights across encoders and arms, and independently resamples test patients as clusters with continuous weights (Rubin, 1981), carrying each patient’s weight to every occurrence across arms, encoders, and splits; BRACS patients with several labels stay one cluster. No replicate lacked support for a class. Intervals are conditional on these three splits and the available test patients. All estimates carry 95% percentile intervals. For the two primary contrasts, two-sided 97.5% intervals give Bonferroni family coverage of at least 95%, and they decide the outcome before any pattern is discussed: an interval that excludes zero resolves the direction; one wholly below -1 or above $+1$ point resolves a practically meaningful change; one wholly inside $[-1, +1]$ supports practical similarity; any other interval is unresolved. Secondary metrics and classwise patterns carry no significance claims.

3 Results

All 840 selected probes ($2 \text{ datasets} \times 2 \text{ encoders} \times 3 \text{ splits} \times 10 \text{ draws} \times 7 \text{ arms}$) passed the completeness and provenance audit, and every selected fit converged. No bootstrap replicate lacked test support for a class. The Virchow2 arms reproduce the earlier estimates on fresh training draws: D_R at $\rho = 100$ is 3.15 points on TCGA-UT against 2.91 before, and 7.86 points on BRACS against 7.65 (Queisler, 2026g,f). The decomposition identity $\delta_R = \delta_P + \delta_S + \delta_I$ holds exactly by construction.

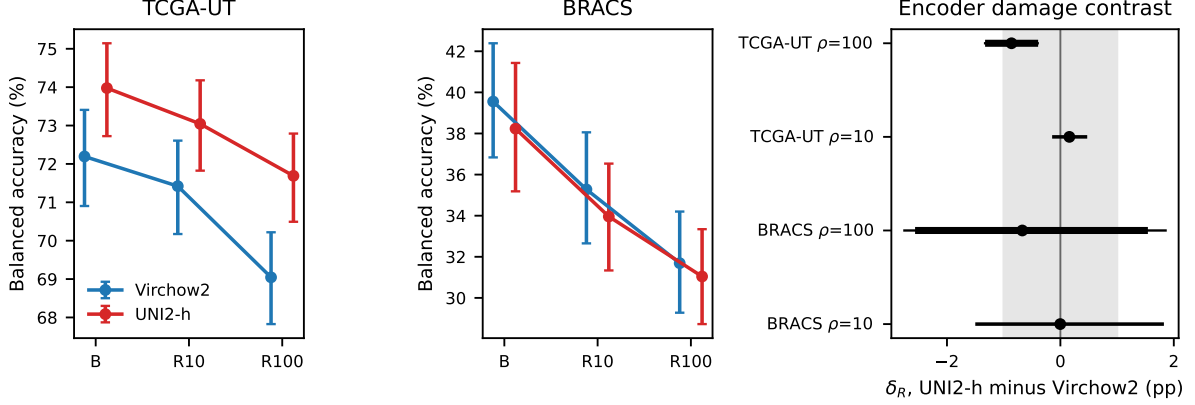


Figure 1: UNI2-h reduces the $\rho = 100$ damage on TCGA-UT while starting from a higher balanced accuracy; on BRACS neither the accuracy levels nor the damage contrast are resolved. Left and centre: test patient-macro balanced accuracy of the balanced arm B and the ratio arms R10 and R100, with paired 95% intervals. Right: encoder contrast δ_R of the damage D_R ; negative values mean less damage with UNI2-h. Thick bars are 95% intervals, the thin bar on each $\rho = 100$ row is the primary two-sided 97.5% interval, and the grey band marks the ± 1 -point practical threshold.

3.1 Balanced and imbalanced accuracy

Absolute accuracy comes first, so that a damage contrast is not read through a floor effect. On TCGA-UT, UNI2-h is more accurate than Virchow2 under balanced training, by 1.78 points (95% interval [1.26, 2.30]), and the advantage grows to 2.64 points [2.12, 3.17] at $\rho = 100$ (Figure 1, left). On BRACS, UNI2-h is 1.32 points less accurate under balanced training, with an interval of $[-3.08, 0.63]$ that includes zero, and 0.65 points $[-1.75, 0.36]$ less accurate at $\rho = 100$. Neither BRACS difference is resolved. Both encoders stay near 38 to 40 percent balanced accuracy on seven BRACS classes and near 72 to 74 percent on thirty TCGA-UT classes, so the dataset, not the encoder, sets the accuracy level.

3.2 Primary damage contrast

On TCGA-UT, UNI2-h loses 2.29 points at $\rho = 100$ and Virchow2 3.15 points. The primary contrast is $\delta_R = -0.86$ points with a 97.5% interval of $[-1.33, -0.40]$. The interval excludes zero, so the direction is resolved: UNI2-h loses less balanced accuracy under this imbalance. It neither lies wholly below -1 point nor wholly inside $[-1, +1]$, so the decision rule leaves open whether the reduction is practically meaningful. All three splits agree in sign and size, with observed contrasts between -0.93 and -0.81 points (Table 5). Because UNI2-h is also the more accurate encoder at both B and R100, the smaller damage is not an artefact of a lower baseline.

On BRACS, UNI2-h loses 7.19 points and Virchow2 7.86 points. The contrast is $\delta_R = -0.67$ points with a 97.5% interval of $[-2.75, 1.85]$, which includes zero and both practical thresholds: the BRACS outcome is unresolved. The splits disagree, from $+0.79$ to -3.13 points, and the draw standard deviation of D_R within a split is 2.3 to 3.9 points on either encoder, so this design cannot distinguish a moderate BRACS reduction from none. At the moderate ratio $\rho = 10$, the damage is almost identical for both encoders: $\delta_R = 0.16$ $[-0.11, 0.44]$ on TCGA-UT and -0.00 $[-1.47, 1.79]$ on BRACS (Table 3).

	Tuned λ per arm (primary)			B-selected λ (sensitivity)		
	Virchow2	UNI2-h	UNI2-h – Virchow2	Virchow2	UNI2-h	Diff.
<i>TCGA-UT</i>						
D_P	1.31 [1.04, 1.59]	1.55 [1.22, 1.89]	0.24 [−0.06, 0.54]	7.93	14.95	7.03
D_S	0.67 [0.49, 0.85]	0.31 [0.16, 0.46]	−0.37 [−0.55, −0.18]	0.74	0.30	−0.44
I	1.17 [0.96, 1.38]	0.43 [0.23, 0.63]	−0.74 [−0.99, −0.50]	0.23	0.36	0.14
D_R	3.15 [2.75, 3.55]	2.29 [1.88, 2.69]	−0.86 [−1.27, −0.45]	8.89	15.62	6.73
<i>BRACS</i>						
D_P	4.51 [3.23, 5.86]	3.89 [2.41, 5.62]	−0.62 [−2.31, 1.37]	11.52	14.94	3.42
D_S	3.76 [2.59, 4.96]	2.09 [0.72, 3.43]	−1.67 [−2.92, −0.52]	4.09	2.58	−1.51
I	−0.41 [−1.77, 0.95]	1.21 [−0.16, 2.73]	1.62 [0.15, 3.21]	−3.59	−2.23	1.36
D_R	7.86 [6.15, 9.63]	7.19 [5.23, 9.32]	−0.67 [−2.50, 1.48]	12.01	15.28	3.27

Table 1: On both datasets UNI2-h carries less support-only damage under either regularization rule, whereas the total and the interaction contrasts depend on how λ is chosen. Damage components at $\rho = 100$ in balanced-accuracy points, defined in Equation (1), with paired 95% intervals; the primary contrast’s 97.5% interval is given in the text. The right block re-evaluates each encoder’s stored P100, S100, and R100 candidates at that encoder’s own B-selected λ , per split and draw; it has no intervals.

3.3 Which operational channel changed

Table 1 splits the $\rho = 100$ damage into its prior, support, and interaction parts. On TCGA-UT the prior-only damage is not reduced: $\delta_P = 0.24 [−0.06, 0.54]$. The reduction comes from the support-only damage, $\delta_S = −0.37 [−0.55, −0.18]$, and above all from the interaction, $\delta_I = −0.74 [−0.99, −0.50]$: on Virchow2 the prior and support channels compound by 1.17 points beyond their sum, on UNI2-h by only 0.43 points. On BRACS, UNI2-h again carries less support-only damage, $\delta_S = −1.67 [−2.92, −0.52]$, but a larger interaction, $\delta_I = 1.62 [0.15, 3.21]$, cancels most of that reduction, and the prior contrast $\delta_P = −0.62 [−2.31, 1.37]$ is unresolved.

The decomposition must also survive the choice of regularization. When every arm is held at its encoder’s B-selected λ , both encoders lose far more accuracy, mainly through the prior arm, and the ordering of total damage reverses: UNI2-h then loses 6.73 points more than Virchow2 on TCGA-UT and 3.27 points more on BRACS (Table 1, right). The TCGA-UT interaction reduction also disappears under that rule. Only the support-channel contrast keeps its sign under both rules on both datasets, at −0.44 and −1.51 points under the fixed rule. The tuned primary result is therefore a statement about the full pipeline, in which each encoder’s probe re-selects its penalty for each imbalanced training set; it is not a statement that UNI2-h’s features resist a shifted prior at a common regularization strength. The penalty scale is not comparable across encoders either: UNI2-h’s validation features have mean Euclidean norms of 16.9 (TCGA-UT) and 15.3 (BRACS), against 49.4 and 46.4 for Virchow2 (Table 7), so one numerical λ regularizes the two encoders to a different degree.

3.4 Probability quality

Table 2 reports macro NLL and ECE at B and R100. Raw UNI2-h probabilities are less well calibrated under balanced training on both datasets: ECE is about 10 points higher, and the fitted temperatures of 0.65 (TCGA-UT) and 0.56 (BRACS), below one, show that the raw probabilities are underconfident. Temperature scaling removes most of that gap. After scaling, UNI2-h has the lower NLL on TCGA-UT at both B (−0.06 nats [−0.08, −0.04]) and R100, while on BRACS its balanced NLL remains 0.06 nats [0.02, 0.10] higher. The imbalance damage in scaled NLL is slightly smaller for UNI2-h on both datasets, by 0.03 nats [−0.04, −0.02] on TCGA-UT and 0.08 nats [−0.11, −0.04] on BRACS. The large reduction in raw ECE damage on TCGA-UT, −7.72 points [−11.01, −4.35], mostly reflects UNI2-h’s poorly calibrated balanced

	B		R100		UNI2-h – Virchow2	
	Virchow2	UNI2-h	Virchow2	UNI2-h	at B	in damage R100–B
<i>TCGA-UT</i>						
NLL raw	0.95	1.02	1.53	1.82	0.08 [0.03, 0.12]	0.21 [0.01, 0.40]
NLL scaled	0.92	0.86	1.05	0.96	−0.06 [−0.08, −0.04]	−0.03 [−0.04, −0.02]
ECE raw	5.9	16.0	17.1	19.5	10.11 [7.11, 13.04]	−7.72 [−11.01, −4.35]
ECE scaled	1.4	2.2	2.0	3.4	0.78 [0.35, 1.10]	0.57 [−0.06, 1.15]
<i>BRACS</i>						
NLL raw	1.55	1.68	8.23	9.20	0.12 [0.08, 0.17]	0.85 [−1.08, 2.87]
NLL scaled	1.56	1.62	1.77	1.75	0.06 [0.02, 0.10]	−0.08 [−0.11, −0.04]
ECE raw	9.2	19.2	39.1	45.4	9.98 [3.67, 14.37]	−3.70 [−9.78, 4.30]
ECE scaled	5.6	8.0	9.7	10.2	2.38 [0.02, 4.75]	−1.91 [−5.09, 1.28]

Table 2: Raw UNI2-h probabilities start less well calibrated, and after temperature scaling the imbalance damage in probability quality is similar for both encoders. Macro NLL in nats and ECE in percentage points, raw and after validation temperature scaling, with paired 95% intervals for the encoder contrasts. The damage contrast compares each encoder’s R100-minus-B change. Ratio 10 is in Table 4.

starting point, not better calibration under imbalance: after scaling, the damage contrast in ECE is 0.57 points [−0.06, 1.15].

3.5 Where the damage lands

On TCGA-UT the lower damage shows in the tail. The tail third of classes, by assigned rank, loses 12.5 points of recall on Virchow2 at $\rho = 100$ but 7.1 points on UNI2-h, while the head third gains 5.3 and 2.3 points (Table 6). On BRACS both encoders collapse the tail third to about 9 to 10 percent recall, from about 41 percent under balanced training, so the tail offers no room for an encoder difference. Per class, averaged over the ranks each class visits across draws, 22 of the 30 TCGA-UT classes lose less recall with UNI2-h; on BRACS 4 of 7 do, with differences of several points in both directions (Figure 2). The validation diagnostics are consistent with this split between datasets but do not explain it. On TCGA-UT, UNI2-h has the higher validation balanced accuracy (74.2 against 72.3 percent) and a larger standardized validation margin (0.72 against 0.64), where the margin is the true-class logit minus the strongest competing logit, divided by that model’s validation margin standard deviation. On BRACS, UNI2-h has the lower validation accuracy (42.2 against 43.4 percent) and a margin near zero (0.05 against 0.15). The most confused class pairs are the same for both encoders: colon against rectal adenocarcinoma on TCGA-UT, and ADH and DCIS predicted as FEA and PB against N on BRACS. These comparisons remain exploratory, because the encoders differ in architecture, dimension, training data, and pooling at the same time.

4 Discussion

Did balanced performance change? On TCGA-UT, yes: UNI2-h is 1.78 points more accurate under balanced training. On BRACS the difference, −1.32 points, is unresolved.

Did imbalance damage change? On TCGA-UT the direction is resolved: under the tuned pipeline UNI2-h loses 0.86 points less balanced accuracy at $\rho = 100$, consistently across the three splits, while also being more accurate at both ends. Whether that reduction is practically meaningful is unresolved, because the primary interval straddles the one-point threshold. On BRACS the primary contrast is unresolved. It does not support practical similarity either, because its interval extends beyond both thresholds. At $\rho = 10$ the two encoders lose almost the same accuracy on both datasets, so any encoder difference emerges only at severe imbalance.

The large-damage regime of BRACS, a 7-to-8-point loss with a collapsed tail, is therefore not a Virchow2 peculiarity: a second encoder with different pretraining data and pooling shows the same regime.

Which operational channel changed? The only contrast that holds its sign on both datasets and under both regularization rules is a smaller support-only damage for UNI2-h: with a uniform class prior, redistributing patches toward the head costs UNI2-h less accuracy. The prior-only damage is not reduced under tuned regularization, and the interaction behaves differently on the two datasets: smaller on TCGA-UT, larger on BRACS, where it absorbs the support reduction. Under a fixed, B-selected penalty the prior damage of UNI2-h becomes much larger, which shows that the tuned result depends on each imbalanced probe re-selecting a weaker penalty. These are statements about the operational prior and support channels of the fitted pipeline. They do not identify a biological or representational mechanism; establishing feature geometry as a causal mediator would need a separate intervention that changes geometry while holding the encoder fixed.

4.1 Limitations

The study compares two encoders with one linear readout on fixed patients and fixed splits; the results do not generalize to foundation models as a category. The two datasets use different support budgets, so the BRACS–TCGA-UT difference is descriptive. The encoders differ in native embedding width, transform, pooling, and pretraining data at once, and the report cannot attribute a contrast to any one of them. The public descriptions of both encoders’ pretraining data were not audited against the TCGA-UT and BRACS source slides beyond reading the UNI2-h model card, so undocumented overlap cannot be excluded. The intervals condition on the available test patients, which earlier studies in this series have already examined. The B-selected- λ sensitivity has point estimates only.

4.2 Protocol deviations

Four points depart from the frozen plan. First, the engineering pilot selected the lower grid endpoint $\lambda = 10^{-8}$ once, for UNI2-h’s R100 arm on TCGA-UT, on a validation plateau (0.7025388 against 0.7025316 at 10^{-7}); the grid was not expanded, contrary to the protocol’s expansion rule. Second, 20 of the 840 main selections landed on a grid endpoint: four UNI2-h fits on TCGA-UT (R10 and R100, at 10^{-8}); two Virchow2 R100 fits on BRACS at 10^{-8} ; and fourteen UNI2-h fits on BRACS, eight at 10^{-8} (P10, P100, R100) and six at 10^2 (B, S10, S100). The protocol prescribes adjacent-decade diagnostic refits for such cells; they were not run, and every estimate above uses the frozen grid. Third, the phase-two input audit mislabeled test-only patch counts as validation-plus-test counts; a signed amendment corrected the counts without changing the schedule, model, grid, or inference plan. Fourth, the per-split contrasts, the paired probability-quality contrasts, the per-class recall, and the validation diagnostics were computed after the locked analysis, by a separate read-only pass over the same accepted fit records; no test result informed any selection.

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A Supplementary results

This appendix collects the secondary displays referred to in the results: the moderate ratio $\rho = 10$, per-split effects, rank-group recall, per-class recall, and the validation-only probe diagnostics.

	Virchow2	UNI2-h	UNI2-h – Virchow2
<i>TCGA-UT</i>			
D_P	0.59 [0.40, 0.80]	0.72 [0.43, 1.01]	0.13 [−0.13, 0.38]
D_S	0.02 [−0.07, 0.12]	−0.01 [−0.07, 0.05]	−0.03 [−0.14, 0.07]
I	0.16 [0.07, 0.25]	0.22 [0.13, 0.32]	0.07 [−0.07, 0.20]
D_R	0.77 [0.55, 1.00]	0.93 [0.64, 1.23]	0.16 [−0.11, 0.44]
<i>BRACS</i>			
D_P	3.21 [2.07, 4.39]	3.25 [1.82, 4.94]	0.04 [−1.42, 1.80]
D_S	1.08 [0.33, 1.81]	0.43 [−0.80, 1.38]	−0.65 [−1.73, 0.33]
I	−0.01 [−0.83, 0.82]	0.60 [−0.62, 2.07]	0.60 [−0.76, 2.12]
D_R	4.28 [3.03, 5.61]	4.28 [2.61, 6.17]	−0.00 [−1.47, 1.79]

Table 3: At $\rho = 10$ the two encoders lose almost the same accuracy on both datasets. Damage components in balanced-accuracy points with paired 95% intervals.

	B		R10		UNI2-h – Virchow2	
	Virchow2	UNI2-h	Virchow2	UNI2-h	at B	in damage R10–B
<i>TCGA-UT</i>						
NLL raw	0.95	1.02	0.98	0.99	0.08 [0.03, 0.12]	−0.06 [−0.14, 0.04]
NLL scaled	0.92	0.86	0.95	0.91	−0.06 [−0.08, −0.04]	0.02 [0.01, 0.03]
ECE raw	5.9	16.0	6.2	6.6	10.11 [7.11, 13.04]	−9.74 [−12.39, −6.88]
ECE scaled	1.4	2.2	1.6	2.4	0.78 [0.35, 1.10]	0.04 [−0.21, 0.30]
<i>BRACS</i>						
NLL raw	1.55	1.68	2.67	4.40	0.12 [0.08, 0.17]	1.61 [0.52, 2.82]
NLL scaled	1.56	1.62	1.66	1.69	0.06 [0.02, 0.10]	−0.03 [−0.06, −0.00]
ECE raw	9.2	19.2	19.2	25.5	9.98 [3.67, 14.37]	−3.65 [−10.02, 4.81]
ECE scaled	5.6	8.0	7.3	8.9	2.38 [0.02, 4.75]	−0.72 [−3.68, 1.58]

Table 4: Probability quality at $\rho = 10$, in the layout of Table 2. UNI2-h’s raw ECE on TCGA-UT falls from B to R10 because its raw balanced probabilities are underconfident.

Dataset	Split	BA of B (%)		$D_R(100)$ (draw SD)		δ_R
		Virchow2	UNI2-h	Virchow2	UNI2-h	
TCGA-UT	0	72.0	73.8	3.44 (0.94)	2.51 (1.01)	-0.93
	1	72.0	74.0	3.11 (0.80)	2.27 (0.62)	-0.84
	2	72.6	74.2	2.89 (0.72)	2.08 (0.67)	-0.81
BRACS	0	39.7	39.2	6.64 (3.64)	6.96 (3.92)	0.33
	1	39.3	39.1	8.79 (2.28)	9.58 (3.21)	0.79
	2	39.7	36.4	8.17 (3.60)	5.04 (2.57)	-3.13

Table 5: The TCGA-UT damage reduction appears in every split; the BRACS splits disagree. Observed per-split means over ten draws, with the standard deviation of D_R across draws in parentheses.

Dataset	Third	Virchow2			UNI2-h		
		B	R100	Change	B	R100	Change
TCGA-UT	head	71.5	76.8	5.3	73.3	75.6	2.3
	body	72.7	70.5	-2.2	74.2	72.1	-2.1
	tail	72.4	59.9	-12.5	74.4	67.3	-7.1
BRACS	head	39.0	51.6	12.6	37.7	50.0	12.2
	body	38.6	23.7	-14.9	35.8	24.6	-11.2
	tail	41.4	9.8	-31.6	41.5	9.2	-32.3

Table 6: On TCGA-UT UNI2-h keeps more tail recall at $\rho = 100$; on BRACS both encoders collapse the tail. Mean test patient-macro recall (%) of the head, body, and tail thirds of classes by their assigned rank in each draw.

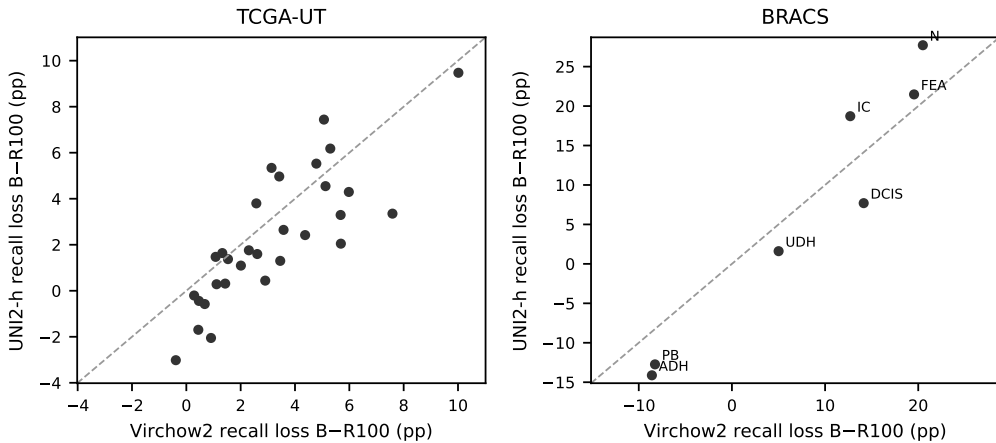


Figure 2: Most TCGA-UT classes lose less recall with UNI2-h; BRACS classes scatter on both sides of the diagonal. Each point is one class’s recall under B minus its recall under R100, averaged over all 30 draws and therefore over the ranks the class visits; points below the dashed diagonal lose less with UNI2-h.

Dataset	Encoder	Dim.	Norm	B				R100				Edge
				Val. BA	Margin	Std.	T	Val. BA	Margin	Std.	T	
TCGA-UT	Virchow2	2560	49.4	72.3	1.81	0.64	0.93	69.2	3.67	0.54	2.17	0
	UNI2-h	1536	16.9	74.2	1.66	0.72	0.65	71.6	7.98	0.61	3.13	4
BRACS	Virchow2	2560	46.4	43.4	0.19	0.15	0.78	35.7	-3.62	-0.19	15.39	2
	UNI2-h	1536	15.3	42.2	0.02	0.05	0.56	33.8	-4.81	-0.25	16.59	14

Table 7: Validation-only probe diagnostics. Norm is the mean Euclidean norm of the native validation features. Val. BA is the selected probe’s validation patient-macro balanced accuracy (%). Margin is the mean raw logit margin of validation patches, the true-class logit minus the strongest competing logit; Std. divides each fit’s mean margin by that fit’s margin standard deviation (no fit had zero variance). T is the mean fitted temperature. Edge counts selections at a grid endpoint over all seven arms and 30 fits.